

The three *standard* architectures = two binary switches

self-attention

cross-attention

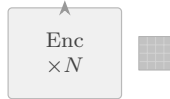
absent

present

bidirectional

Encoder-only

pooled embedding



tokens

BERT

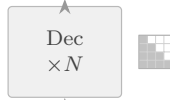
non-standard

bidirectional self-attention
is incompatible with the
autoregressive generation
cross-attention serves

causal

Decoder-only

next token

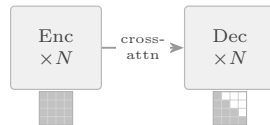


prompt

GPT, LLaMA

Encoder-decoder

next token



T5, original transformer

Row = the self-attention of the *generating* stack — encoder-decoder is causal because its decoder generates.
Three of the four settings are standard; a bidirectional generator with cross-attention (top-right) is unused.



attend all positions (bidirectional)



attend earlier positions only (causal)